

Phu Vinh Nguyen

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EDUCATION

Northeastern University Boston, MA
Certificate in Bioinformatics, concentration in Software Development May 2025
Relevant Coursework: Python for Bioinformatics, Genomics in Bioinformatics, Algorithms

Uniwersytet Jagielloński Krakow, PL
Master's in Molecular Biotechnology Jul 2021

University of California, Irvine Irvine, CA
Bachelor of Science in Pharmaceutical Sciences Sept 2019

WORK EXPERIENCE

Automation Engineer | Developer, Pharma - Sanofi Pasteur Jun 2025 – Present
Automation & Data Engineering Team

- Developed a form help request desktop app using React and Electron to standardize researcher form submissions and automate internal ticket creation via SMTP email integration.
- Create, test, and deploy methods to liquid handler robots

Automation Engineer | Developer, Biotech - Foundation Medicine Inc Nov 2024 – Apr 2025
Assay Development, Automation Engineering Team

- Developed and launched a Python-GUI app using CustomTkinter to provide a clean interface for backend automation scripts to enable non-coders to perform tasks independently.
- Built a React troubleshooting app for Hamilton robot devices that display the metadata from specific objects to help cross-functional teams verify dependencies and debug production systems.
- Developed barcode scanning systems and ETL workflows to integrate sample data into Sapio LIMS (laboratory information management systems) to automate sample tracking.

Lab Automation Engineer, Pharma - Sanofi Pasteur Oct 2022 – Nov 2024
Automation Engineering Team

- Wrote validation protocols and step-by-step instructions for new users and trained them on the use of automated equipment.
- Optimized automated high-throughput methods on Hamilton platforms, reducing assay processing times by up to 30% across workflows including fluorescent-based assays and DNA extraction.
- Delivered biweekly presentations to key stakeholders, highlighting automation methods, technical challenges, implemented solutions, and supporting data that improved assay performance and usability.

Senior Research Associate, Startup - Dermibiont Inc Oct 2021 – Sept 2022
Clinical Diagnostic Team

- Built data pipeline analysis for Oxford Nanopore Technology (ONT) using open source bioinformatic tools involving Guppy basecalling and quality control filtering.
- Collaborated with external bioinformatics teams to troubleshoot data pipelines and optimize QC logic for nanopore high-throughput sequencing workflows using Docker.
- Automated DNA extraction protocols on OpenTrons using Python and R for data visualization.

PROJECTS

Personal Website

- Built a react portfolio web app to showcase my professional experience, projects, and skills.

SKILLS

Languages: JavaScript, HTML, CSS, Bash/Shell, Python, SQL
Frameworks and Libraries: React.js, CustomTkinter, Express, Electron.js
Technical Tools: Git, Jira, HPC, User-level LIMS, Docker